SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***GAYATHRI N.P (192421362)*** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Gayathri N P
Signature of the student:

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails.

1. Perceptron Model (5 Marks)

Given:

- Study Hours (x1) = 2
- Attendance (x2) = 1
- Weight (w1) = 0.6
- Weight (w2) = 0.4
- Bias (b) = -0.5

Formula:

$$\text{Net Input} = (w1 \times x1) + (w2 \times x2) + b$$

Calculation:

$$\text{Net Input} = (0.6 \times 2) + (0.4 \times 1) + (-0.5)$$

$$= 1.2 + 0.4 - 0.5$$

$$= 1.1$$

Step Activation Function:

- If Net Input ≥ 0, Output = 1
- Otherwise, Output = 0

Since $1.1 \geq 0$,

Output = 1

Conclusion:

The perceptron classifies the student as **Pass**.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training.

Given:

- Initial Weight (w) = 0.5
- Learning Rate (η) = 0.1
- Input (x) = 2
- Actual Output = 0.6
- Target Output = 1

Step 1: Calculate Error

Error = Target Output - Actual Output

= 1 - 0.6

= 0.4

Step 2: Delta Rule

$\Delta w = \eta \times \text{Error} \times \text{Input}$

$= 0.1 \times 0.4 \times 2$

= 0.08

Step 3: Update Weight

New Weight = Old Weight + Δw

= 0.5 + 0.08

= 0.58

Final Answer:

- Error = 0.4
- Weight Update = 0.08
- New Weight = 0.58

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors.

Given:

- True Positive (TP) = 180
- True Negative (TN) = 150
- False Positive (FP) = 20
- False Negative (FN) = 50

Formula 1: Accuracy

Accuracy = (TP + TN) / (TP + TN + FP + FN)

= (180 + 150) / (180 + 150 + 20 + 50)

= 330 / 400

= 0.825 = 82.5%

Formula 2: Precision

Precision = TP / (TP + FP)

= 180 / (180 + 20)

= 180 / 200

= 0.90 = 90%

Formula 3: Sensitivity (Recall)

Sensitivity = TP / (TP + FN)

= 180 / (180 + 50)

= 180 / 230

= 0.7826 = 78.26%

Formula 4: Specificity

Specificity = TN / (TN + FP)

= 150 / (150 + 20)

= 150 / 170

= 0.8824 = 88.24%

Final Results:

- Accuracy = 82.5%
- Precision = 90%
- Sensitivity (Recall) = 78.26%

- Specificity = 88.24%

Conclusion:

The model has good accuracy and precision. However, the recall is 78.26%, meaning some diseased patients are classified as healthy (false negatives). In medical diagnosis, reducing false negatives is important, so improving recall would make the model more reliable.

4.A Genetic Algorithm is applied to maximize the function $f(x) = x^2$ using an initial population consisting of the values 2, 4, 6, and 8. Determine the fitness value for each individual in the population and select the best candidates based on their fitness scores. Perform a crossover operation by exchanging the least significant bits of the selected individuals to generate new offspring. Additionally, explain the role of mutation in improving the diversity and effectiveness of the solution in Genetic Algorithms.

Given:

Function:

$f(x) = x^2$

Initial Population:

2, 4, 6, 8

Step 1: Calculate Fitness

$x = 2 \rightarrow f(x) = 2^2 = 4$

$x = 4 \rightarrow f(x) = 4^2 = 16$

$x = 6 \rightarrow f(x) = 6^2 = 36$

$x = 8 \rightarrow f(x) = 8^2 = 64$

Step 2: Select Best Individuals

Best candidates:

- x = 8 (Fitness = 64)
- x = 6 (Fitness = 36)

Step 3: Binary Representation

6 = 0110

8 = 1000

Exchange the least significant bit (LSB):

Parent 1 = 0110

Parent 2 = 1000

After swapping the last bit:

Offspring 1 = 0110 = 6

Offspring 2 = 1000 = 8

Since both last bits are 0, the offspring remain unchanged.

Role of Mutation:

- Maintains diversity in the population.
- Prevents premature convergence.
- Helps escape local optimum solutions.
- Improves the chance of finding the global optimum.

5 .During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model.

Given:

- Training Accuracy: 60% → 95%
- Validation Accuracy: 58% → 72% → 68%

Issue Identified:

The model is suffering from **Overfitting**.

Reason:

Overfitting occurs when the neural network learns the training data too well, including noise and unnecessary details. As a result, training accuracy keeps increasing while validation accuracy starts decreasing.

Techniques to Reduce Overfitting:

1. **Dropout**
    - Randomly deactivates neurons during training.
    - Prevents the network from depending too much on specific neurons.
2. **Early Stopping**
    - Stops training when validation accuracy begins to decrease.
    - Prevents unnecessary training and reduces overfitting.

Significance of Validation Accuracy:

- Measures performance on unseen data.
- Evaluates how well the model generalizes.

- Helps detect overfitting and underfitting.
- Assists in selecting the best-performing model.

Conclusion:

Since training accuracy increases to 95% while validation accuracy decreases from 72% to 68%, the model is **overfitting**. Using **Dropout** and **Early Stopping** can improve the model's performance on new data.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand